

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0657

Hawthorne 138 kV
April 20, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0657. The Interconnection Customer proposes a 110.5 MW wind facility interconnecting at the Hawthorne 138 kV substation in the New Jersey area. The request was received on July 29, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 245 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,735,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Colorado to Yahara River 138 kV	Sne - Sansum N Elec to Multitrade Of Pittsylvania Lp Substation 138 kV	214.5	235	8.0
West Point Municipal to Yahara Riv 138 kV L	Russell Energy Center to Fair Lawn 138 kV	197.3	244	29.4
Oleta to Montclair Cogen Facility 138 kV	Sne - Sansum N Elec to Multitrade Of Pittsylvania Lp 138 kV	156.2	186	6.8
Colorado to Oleta 138 kV L	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec Substation 138 kV L	153.9	216	6.8
Russell Energy Ctr to Fair Lawn Substation 138 kV	W Pt Municipal to Yahara River 138 kV	150.5	223	31.5
Russell Energy Ctr to Hawthorne 138 kV	Walnut-Engle Tap to Cabinet Gorge 138 kV	146.5	174	73.1
West Point Municipal to Cabinet Gorge 138 kV line	Russell Energy Center to Fair Lawn 138 kV	138.3	196	44.4
Fair Lawn to Manor 138 kV	W Pt Municipal to Yahara River 138 kV	127.8	206	31.5
Colorado to Nietos 138 kV L	Russell Energy Center to Fair Lawn 138 kV	122.9	172	7.9
Lane to Passaic Vly Water Commission 138 kV line	Colorado to Yahara Riv 138 kV	112.3	214	5.3
Spara to Montclair Cogen Facility 138 kV	Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138 kV line	110.2	223	6.8
Forest Hill to Brownfield City 138 kV	Russell Energy Ctr to Hawthorne 138 kV	109.1	262	13.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Black Hollow to Manor 138 kV L	W Pt Municipal to Yahara Riv 138 kV Line	108.4	222	31.5
Smoky Hill to Forest Hill 138 kV	Russell Energy Center to Hawthorne 138 kV	104.8	259	13.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Colorado to Yahara River 138 kV	Rebuild line with higher-capacity conductor	\$1,096,000	100% (\$1,096,000)
West Point Municipal to Yahara Riv 138 kV L	Rebuild line with higher-capacity conductor	\$1,680,000	100% (\$1,680,000)
Oleta to Montclair Cogen Facility 138 kV	Rebuild line with higher-capacity conductor	\$1,149,000	100% (\$1,149,000)
Colorado to Oleta 138 kV	Rebuild line with higher-capacity conductor	\$1,369,000	100% (\$1,369,000)
Russell Energy Ctr to Fair Lawn Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,587,000	100% (\$1,587,000)
Russell Energy Center to Hawthorne 138 kV	Rebuild line with higher-capacity conductor	\$1,301,000	100% (\$1,301,000)
West Point Municipal to Cabinet Gorge 138 kV line	Rebuild line with higher-capacity conductor	\$3,411,000	100% (\$3,411,000)
Fair Lawn to Manor 138 kV	Rebuild line with higher-capacity conductor	\$3,249,000	100% (\$3,249,000)
Colorado to Nietos 138 kV L	Rebuild line with higher-capacity conductor	\$901,000	100% (\$901,000)
Lane to Passaic Vly Water Commission 138 kV line	Rebuild line with higher-capacity conductor	\$3,125,000	100% (\$3,125,000)
Spadra to Montclair Cogen Facility Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,219,000	100% (\$1,219,000)
Brownfield City to Forest Hill Substation 138 kV	Reconductor line	\$1,323,000	100% (\$1,323,000)
Manor to Black Hollow 138 kV	Reconductor line	\$1,385,000	100% (\$1,385,000)
Smoky Hill to Forest Hill 138 kV	Reconductor line	\$940,000	100% (\$940,000)
	Total allocated to project		\$23,735,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.